

Advanced (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. A group of friends want to start a small gardening project. They plan to plant 3 trees, and 2^x flowers around each tree, with x being the number of days after planting the trees. If they plant 3 trees today, and want to have 48 flowers around each tree after x days, find the value of x using the formula $2^x =$

$\frac{48}{3}$, and show all steps to solve for x . You must use the properties of exponents.

Q10. Explain how the concept of exponents is used in time management when planning a study schedule for an exam. Include examples of how the rules of exponents can be applied to optimize study time, using the formula for compound interest $A = P(1 + r)^n$ where A is the total amount of material studied, P is the principal amount of material, r is the rate of studying, and n is the number of days. You must use mathematical expressions and provide a real-life scenario.

Chef's Tips

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- Tip 1: Encourage students to use the rules of exponents to simplify expressions and solve problems.
- Tip 2: Remind students to apply the rules of exponents in different contexts, such as science, sports, and daily life.
- Tip 3: Emphasize the importance of checking work and using estimation to verify answers.